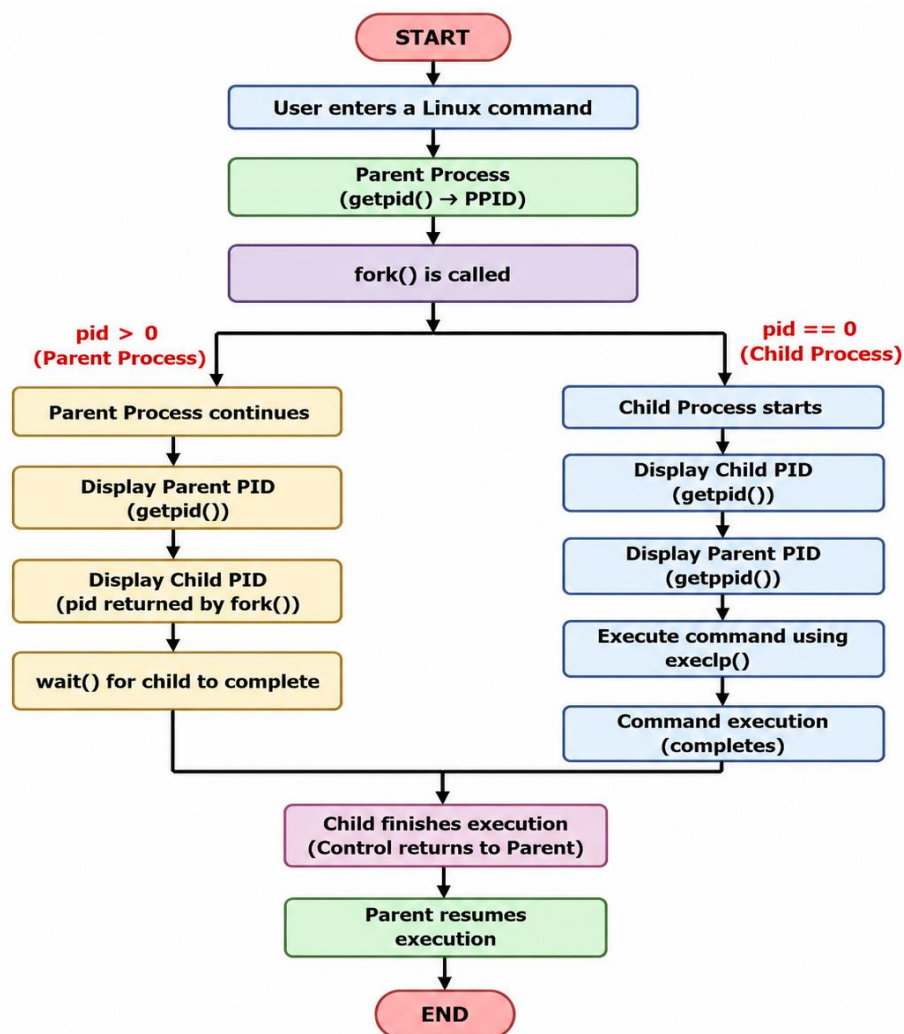


Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using `fork()`. 3. Execute the command in the child process using an appropriate `exec()` system call. 4. Allow the parent process to wait for the child using `wait ()`. 5. Display the Process ID (PID) of both parent and child processes.

- Write a C code to create a process using `fork()` system call.
- Read the user input as shell command.
- Use a system call `exec()` to execute the shell command.
- Use a `wait()` system call for waiting a parent process.
- Display the PIDs of parent and child processes.



Example Output:

Enter a Linux command: ls

Parent Process

Child Process

Parent PID : 67

Child PID : 68

Child PID : 68

Parent PID : 67

Executing command: ls

CE1 FIRST.c File2.c Proces_States.c command_executor filecopy.c
hello.c process_example session_4_a session_4_b.c

CE1.c File1.c First.c Process_States.c command_executor.c fork_example.c.save
my_shell process_example.c session_4_a.c

FIRST File1.txt Proces_States R1 filecopy hello
os-lab process_example.cyes session_4_b

Child process completed.

```
alekhya@AlekhyaReddy: ~  
alekhya@AlekhyaReddy:~$ nano task1.c  
alekhya@AlekhyaReddy:~$ ls  
LICENSE          command_executor.c  nano.9019.save  
Makefile         docs               obj  
OSLab            fork_exec          process  
OSSP             fork_exec.c        process.c  
README.md        fork_program       scripts  
Week2            hello              src  
assets           hello.c            task1.c  
bin              include            task1.c  
command_executor main.c             tests  
alekhya@AlekhyaReddy:~$ gcc task1.c -o task1  
alekhya@AlekhyaReddy:~$ ./task1  
Enter a Linux command: ls  
  
Parent Process  
Parent PID : 9815  
Child PID : 9816  
  
Child Process  
Child PID : 9816  
Parent PID : 9815  
LICENSE          command_executor.c  nano.9019.save  
Makefile         docs               obj  
OSLab            fork_exec          process  
OSSP             fork_exec.c        process.c  
README.md        fork_program       scripts  
Week2            hello              src  
assets           hello.c            task1.c  
bin              include            task1.c  
command_executor main.c             tests  
  
Child process completed.  
alekhya@AlekhyaReddy:~$
```

```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0)
    {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        execlp(command, command, NULL);

        // Executes only if exec fails
        printf("Command execution failed!\n");
        exit(1);
    }
    else
    {
        // Parent Process
    }
}
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Uname -a:

```
alekhya@AlekhyaReddy: ~$ uname -a
Linux AlekhyaReddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
alekhya@AlekhyaReddy: ~$
```

Lscpu:

```
alekhya@Alekhyareddy: ~  
alekhya@Alekhyareddy:~$ uname -a  
Linux Alekhyareddy 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux  
alekhya@Alekhyareddy:~$ lscpu  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:          42 bits physical, 48 bits virtual  
Byte Order:            Little Endian  
CPU(s):                8  
On-line CPU(s) list:   0-7  
Vendor ID:             GenuineIntel  
Model name:            Intel(R) Core(TM) Ultra 7 258V  
CPU family:            6  
Model:                189  
Thread(s) per core:    1  
Core(s) per socket:    8  
Socket(s):             1  
Stepping:              1  
BogoMIPS:              6604.83  
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe  
1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse  
3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3  
dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bml avx2 smep bmi2 erms  
invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpc  
lmulqdq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_lld arch_capabilities  
  
Virtualization features:  
Virtualization:        VT-x  
Hypervisor vendor:     Microsoft  
Virtualization type:   full  
Caches (sum of all):  
L1d:                   384 KiB (8 instances)  
L1i:                   512 KiB (8 instances)  
L2:                    20 MiB (8 instances)  
L3:                    12 MiB (1 instance)  
NUMA:  
NUMA node(s):          1  
NUMA node0 CPU(s):     0-7  
Vulnerabilities:  
Gather data sampling:   Not affected  
Ghostwrite:            Not affected  
Indirect target selection: Not affected  
Itlb multihit:         Not affected  
L1tf:                  Not affected  
Mds:                   Not affected  
Meltdown:              Not affected  
Mmio stale data:       Not affected  
Old microcode:         Not affected  
Reg file data sampling: Not affected  
Retbleed:              Mitigation; Enhanced IBRS  
Spec rstack overflow:  Not affected  
Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl  
Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization  
Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S  
Srbds:                 Not affected  
Tsa:                   Not affected  
Tsx async abort:       Not affected  
Vmscape:               Not affected  
alekhya@Alekhyareddy:~$
```

Ps:

```
alekhya@Alekhyareddy:~$ ps  
  PID TTY          TIME CMD  
 9869 pts/3        00:00:00 bash  
 9902 pts/3        00:00:00 ps  
alekhya@Alekhyareddy:~$
```

Top:

```
9902 pts/3    00:00:00 ps
alekhya@Alekhya:~$ top
top - 05:44:33 up 22:36,  1 user,  load average: 0.04, 0.04, 0.00
Tasks:  29 total,   1 running, 28 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem : 15743.6 total, 15022.7 free,   668.7 used,   251.7 buff/cache
MiB Swap:  4096.0 total,  4096.0 free,    0.0 used. 15074.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21848	13236	9800	S	0.0	0.1	0:04.48	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.25	init-systemd(Ub
6	root	20	0	3220	2144	2008	S	0.0	0.0	0:00.16	init
52	root	19	-1	66828	17980	16836	S	0.0	0.1	0:03.06	systemd-journal
109	root	20	0	25640	6792	5100	S	0.0	0.0	0:26.97	systemd-udev
117	systemd+	20	0	21344	13192	11068	S	0.0	0.1	0:00.50	systemd-resolve
118	systemd+	20	0	91036	7960	6992	S	0.0	0.0	0:01.60	systemd-timesyn
165	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.35	cron
171	message+	20	0	9600	5452	4736	S	0.0	0.0	0:00.59	dbus-daemon
179	root	20	0	17976	8768	7736	S	0.0	0.1	0:00.53	systemd-logind
190	syslog	20	0	222516	5620	4568	S	0.0	0.0	0:00.42	rsyslogd
201	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
214	root	20	0	107036	23276	13820	S	0.0	0.1	0:00.08	unattended-upgr
269	root	20	0	6696	4652	3876	S	0.0	0.0	0:00.01	login
335	alekhya	20	0	20324	11584	9456	S	0.0	0.1	0:00.26	systemd
336	alekhya	20	0	21164	3572	1848	S	0.0	0.0	0:00.00	(sd-pam)
365	alekhya	20	0	6064	5360	3696	S	0.0	0.0	0:00.01	bash
728	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00	SessionLeader
729	root	20	0	3204	1256	1108	S	0.0	0.0	0:00.02	Relay(730)
730	alekhya	20	0	6076	5420	3704	S	0.0	0.0	0:00.04	bash
1682	polkitd	20	0	308172	8104	7204	S	0.0	0.1	0:03.61	polkitd
1810	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00	SessionLeader
1811	root	20	0	3204	1256	1108	S	0.0	0.0	0:00.04	Relay(1814)
1814	alekhya	20	0	6080	5432	3712	S	0.0	0.0	0:00.06	bash
9821	root	20	0	1830360	13436	11208	S	0.0	0.1	0:00.18	wsl-pro-service
9863	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00	SessionLeader
9865	root	20	0	3204	1256	1108	S	0.0	0.0	0:00.01	Relay(9869)
9869	alekhya	20	0	6080	5372	3656	S	0.0	0.0	0:00.03	bash
9904	alekhya	20	0	9400	5800	3560	R	0.0	0.0	0:00.02	top